

Packet Tracer - Connect Devices using Wireless Technologies

Objectives

Part 1: Connect a Laptop to the Office WLAN

Part 2: Connect Devices with Bluetooth Technology

Part 3: Tether a Laptop to Use a Cellular Network via the Smartphone

Background

In this Packet Tracer activity, you will use different wireless technologies to connect end devices in an office. The activity is performed in the Packet Tracer Physical Mode only.

Required Resources

- Latest Packet Tracer version

Instructions

Part 1: Connect a Laptop to the Office WLAN

Step 1: Install a wireless module to a Laptop.

- Click the **Laptop** to open the configuration window.
- Under the **Physical** tab, power off the **Laptop** by clicking the power button.
- Remove the Ethernet module **PT-LAPTOP-NM-1CFE** from the laptop by dragging it from the **Laptop** to the list on the left.
- Insert the wireless module **WPC300N** by dragging it from the list on the left to the **Laptop**.
- Power on the **Laptop**.

Step 2: Connect Laptop to the office WLAN.

- Click the **Desktop** tab and select the **PC Wireless** tool.
- Click the **Connect** tab and wait until the **Employee SSID WLAN** is displayed. Note that you may have to click **Refresh**.
- Click the **Employee SSID** to select it. Click **Connect**.
- Enter **Cisco123** as the pre-shared key and click **Connect**.
- After connecting to the wireless network, close the **PC Wireless** window.

- f. Click the **Config** tab and select **Wireless0** in the left pane to verify in the **IP Configuration** section that the **Laptop** has been assigned an IP address.
- g. Open the **Web Browser** from the **Desktop**. Navigate to **office.srv** to verify that the **Laptop** has connectivity.
- h. Close the **Laptop** window.

Part 2: Connect Devices with Bluetooth Technology

In this part, you will connect a Bluetooth speaker to a tablet installed with a music player via Bluetooth.

Step 1: Enable Bluetooth ports on devices.

- a. Click the **Bluetooth Speaker**.
- b. Click the **Config** tab.
- c. Click **Bluetooth** on the left pane and check that the Port Status is **On**. Note that the speaker is not paired with the **Office Tablet**.

Step 2: Connect Bluetooth devices.

- a. Open the **Office Tablet**.
- b. Click the **Config** tab.
- c. Click **Bluetooth** in the left pane and check the **On** box for Port Status.
- d. Click **Discover** and the **Bluetooth Speaker** device should be discovered.
- e. Select the **Bluetooth Speaker** in the **Devices** list and click **Pair**. The status should change to "Paired, Connected". If prompted for permission to connect, click **Yes**.
- f. To test the Bluetooth connection, click the **Desktop** tab and select **Music Player**. Click **Play/Stop** to start the music.
Note: Make sure your speaker is on.
- g. Click **Play/Stop** again to stop the sound.

Part 3: Tether a Laptop to Use a Cellular Network via the Smartphone

In this part, you will tether a laptop to a smartphone via Bluetooth. The laptop will use the cellular network to access a website.

Step 1: Enable Bluetooth on the Laptop.

- a. Click the **User-Laptop**. Select the **Config** tab.
- b. Click **Bluetooth** on the left panel. Click **On** for the Port Status.
- c. Leave the **User-Laptop** Bluetooth window open.

Step 2: Connect a smartphone to the Cellular network and enable Bluetooth.

- a. Click the **Smartphone** to open the configuration window.
- b. Click the **Config** tab. Check the **On** box for the Cellular Tethering setting in the **Global Settings**.
- c. Click the **3G/4G Cell1** interface. Verify that the **Smartphone** has an IP address from the cellular network.
- d. Click **Bluetooth** in the left pane and check the **On** box for the Port status on the **Smartphone**.

Step 3: Connect Bluetooth devices and tethering to laptop

- a. On the Bluetooth configuration for the **Smartphone**, click **Discover** to search for nearby Bluetooth enabled-devices.

Note: If **Smartphone** does not appear after a while, move **Smartphone** closer to **User-Laptop** and click **Discover** again.

- b. Select **User-Laptop** and click **Pair**. A pop-up window appears and asks for permission. Click **Yes**. The two devices are connected via Bluetooth.
- c. Return to the **User-Laptop**. In the Bluetooth panel of the **Config** tab, highlight **Smartphone** and click **Tether**.
- d. At the bottom pane of the Bluetooth configuration, notice that **User-Laptop** has now obtained an IP address.
- e. To test connectivity, navigate to **office.srv**. Click **Desktop > Web Browser**. Enter **office.srv** in the **URL** field. You can fast forward time to speed up the process.